upGrad

Methodology Report:
Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

DO NOT EDIT ANYTHING ON THIS PAGE

Points to Note:

1. **Without this methodology document, the other parts of your case study will not be evaluated.**
2. This assignment is different from the ones you have solved before. Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand.
   Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. **DO NOT** change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.
- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

Analysis and Visualisation

1. Data Preparation [10 Marks]

1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

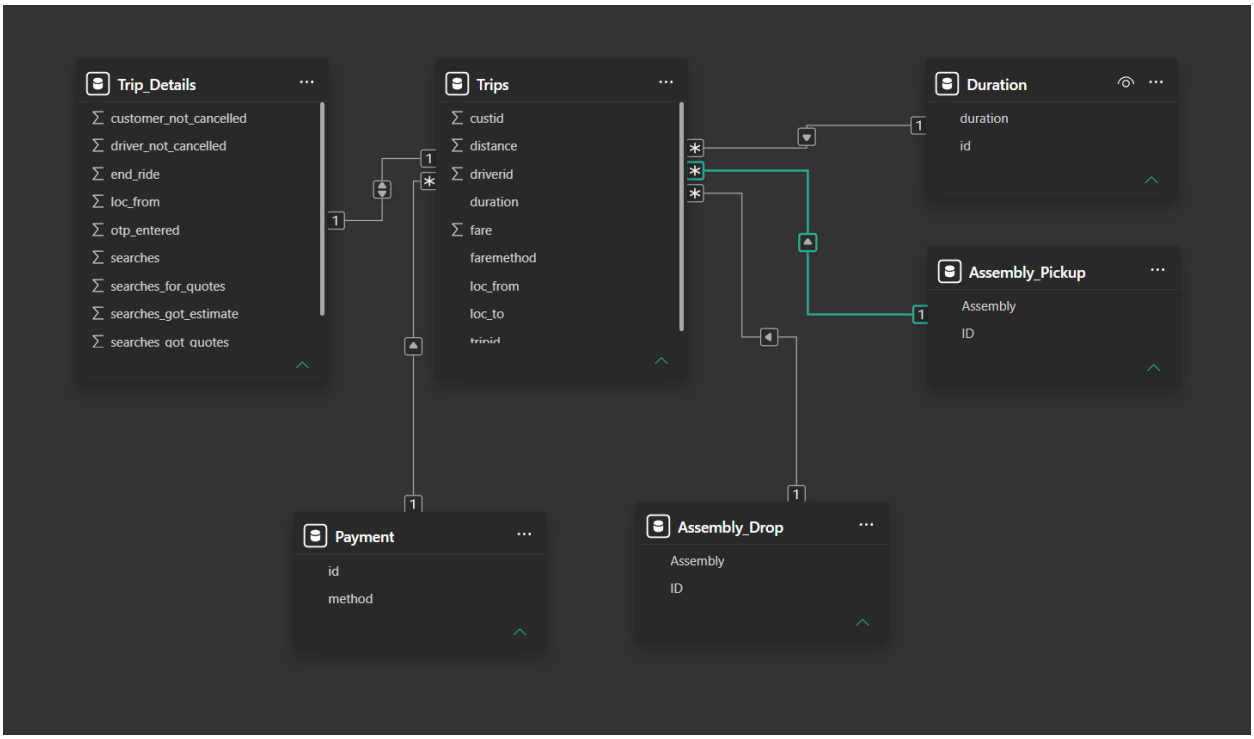
Solution:

The Namma Yatri dataset was imported into Power BI as 5 tables: *Trips*, *Trip_Details*, *Duration*, *Payment*, and *Assembly*.

Joins created:

- Trips.tripid → Trip_Details.tripid (1:1) — links each completed trip to its full search/cancellation funnel
- Trips.duration → Duration.id (many:1) — maps duration codes to readable hour buckets
- Trips.faremethod → Payment.id (many:1) — maps fare method codes to payment names (cash, UPI, etc.)
- Trips.loc_from → Assembly_Pickup.ID (many:1) — maps pickup location to zone name
- Trips.loc_to → Assembly_Drop.ID (many:1) — maps drop location to zone name

Note on Assembly table: Since both loc_from and loc_to in Trips reference the same Assembly zone list, the Assembly table was duplicated into *Assembly_Pickup* and *Assembly_Drop* (role-playing dimension pattern). This avoids Power BI's single-active-relationship limitation and allows pickup and drop zone names to be used simultaneously in visuals.



1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.
- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

Checked the dataset for the usual issues — duplicate trip IDs, missing/negative fares or distances, duplicate zone names, and broken logic in the Trip_Details funnel (e.g. a ride marked "completed" without OTP entry). None of these showed up. No duplicates anywhere, fares range from ₹44–1500 with nothing at zero or negative, distances are all positive, and the funnel columns in Trip_Details (searches → estimate → quote → cancellation flags → OTP → end ride) follow logical order with no row skipping a step it shouldn't.

The one thing that looks like an inconsistency but isn't: Trip_Details has 2,161 rows while Trips has only 983. This is structural, not an error — Trip_Details logs every search attempt, including ones that never turned into a completed ride, so it's expected to be a wider table. This was already accounted for in the join in Section 1.1.

Also verified that all ID columns (tripid, driverid, custid, loc_from, loc_to, duration, faremethod) are correctly typed as whole numbers in Power Query rather than text, since a wrong data type can silently break relationships even when values look fine.

Calculated columns created:

- **Fare per KM** = fare / distance — added to the Trips table to look at pricing/efficiency by zone

2. Exploratory Data Analysis [40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:

Categorical variables:

- Assembly (zone name — Pickup and Drop) — *nominal, no inherent order*
- Payment method (cash, UPI, etc.)
- Duration bucket (e.g. "0-1", "1-2") — stored as numbers but really a labeled time range, not a quantity
- ID fields: tripid, custid, driverid, loc_from, loc_to, faremethod — numeric in the raw data, but they identify a trip/customer/driver/zone rather than measure something, so they're treated as categorical
- Trip_Details funnel flags (searches, searches_got_estimate, searches_for_quotes, searches_got_quotes, customer_not_cancelled, driver_not_cancelled, otp_entered, end_ride) — binary 0/1 values representing yes/no states, used for counts and rates rather than averages

Numerical variables:

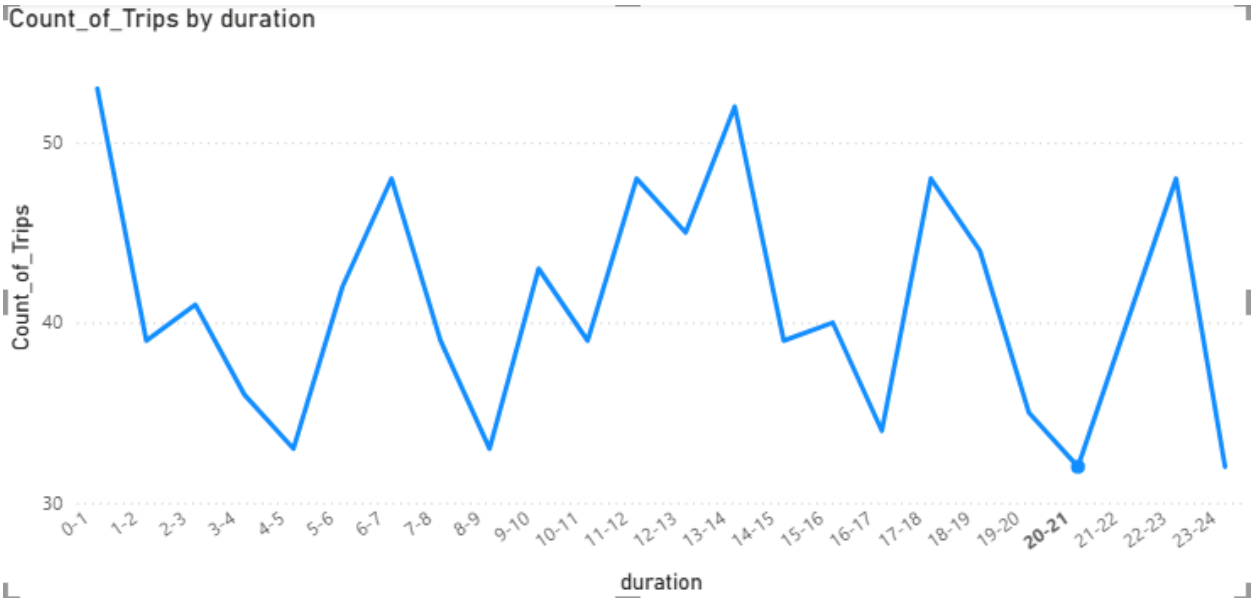
- fare
- distance
- fare per km (calculated column)

Note on Assembly: Assembly is stored as an ID (integer) in the raw table, but after joining it to get the zone name, what's actually being analyzed is the name itself — a categorical/nominal variable with no natural order, used to group and filter rather than calculate.

2.2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:



Demand (count of trips) doesn't follow one simple rush-hour pattern — it's spread across several smaller peaks through the day rather than two clean spikes.

The single highest demand hour is **0-1** (53 trips), followed closely by **13-14** (52 trips). Three other hours tie for the next-highest demand at 48 trips each: **6-7, 11-12, and 17-18** — interestingly all roughly 5-6 hours apart, suggesting a recurring cyclical pattern rather than a one-off morning/evening rush.

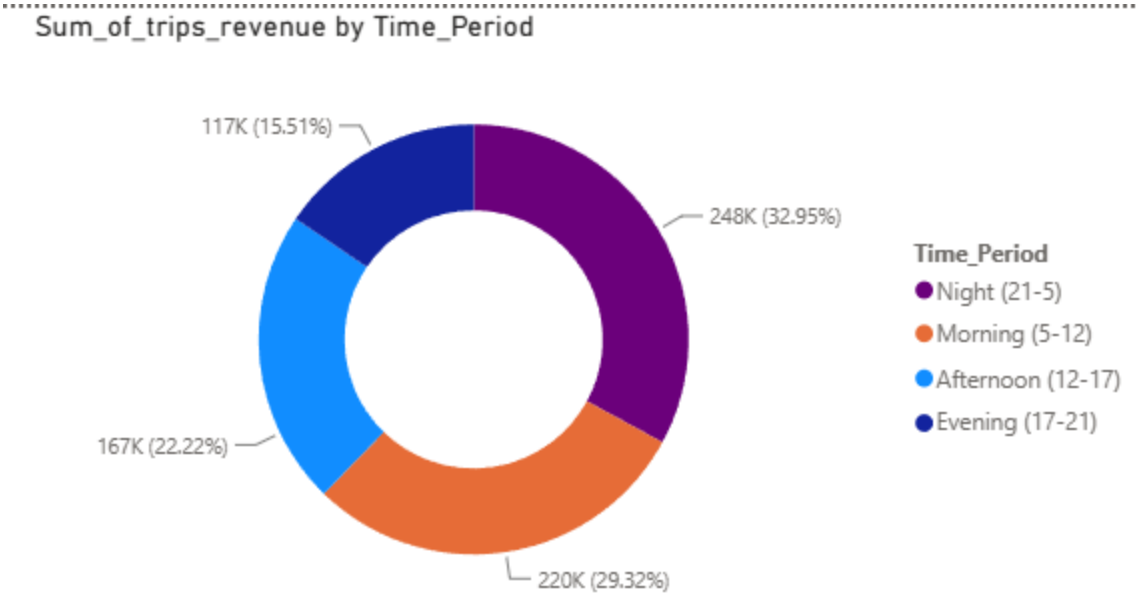
The lowest demand windows are **20-21** and **23-24** (32 trips each), along with **4-5** and **8-9** (33 trips each) — late night and early morning are consistently the quietest periods, alongside a smaller mid-morning dip around 8-9.

Overall, the data shows demand oscillating in a wave pattern — rising and falling roughly every 4-6 hours — rather than a single rush hour story, which is something worth calling out as slightly unusual for a ride-hailing dataset.

2.3. Proportion of Total Revenue from Different Time Periods [3 Marks]

- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

Solution:



Total revenue of ₹7,51,343 was analysed across four time periods using a calculated column (Time_Period) created in the Trips table, grouping the 24 duration buckets into Night (21:00–05:00), Morning (05:00–12:00), Afternoon (12:00–17:00), and Evening (17:00–21:00).

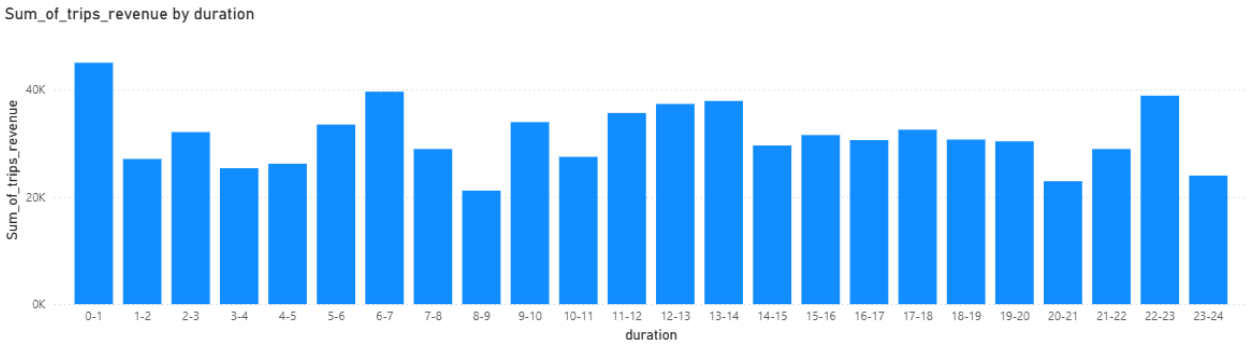
Night (21:00–05:00) contributes the highest share at **32.95%** (₹2,48,000), driven by the midnight demand peak — the 0-1 slot alone is the single highest revenue hour. **Morning (05:00–12:00)** follows at **29.32%** (₹2,20,000), with the 6-7 slot a strong contributor. **Afternoon (12:00–17:00)** accounts for **22.22%** (₹1,67,000), and **Evening (17:00–21:00)** is the weakest at just **15.51%** (₹1,17,000).

The night period dominating revenue despite covering the same number of hours as other periods suggests late-night rides tend to be longer or higher-value trips. Evening being the lowest contributor despite having a demand spike at 17-18 indicates those rides are shorter and cheaper on average — high volume but low revenue per trip, which is worth flagging for pricing strategy.

2.4. Explore the Relationship Between Trip Hour and Revenue [3 Marks]

- Investigate the correlation between trip hour and total fare.
- Explain any trends or patterns that emerge.

Solution:



The bar chart shows total revenue (sum of fare) by trip hour across all 24 duration buckets.

The highest revenue hours are **0-1** (₹45,019), **6-7** (₹39,624), and **22-23** (₹38,880) — these align closely with the peak trip count hours from 2.2, confirming that volume is the primary driver of revenue rather than higher fares at specific times.

The lowest revenue windows are **8-9** (₹21,174), **20-21** (₹22,936), and **23-24** (₹23,979) — again consistent with the low-demand hours identified earlier.

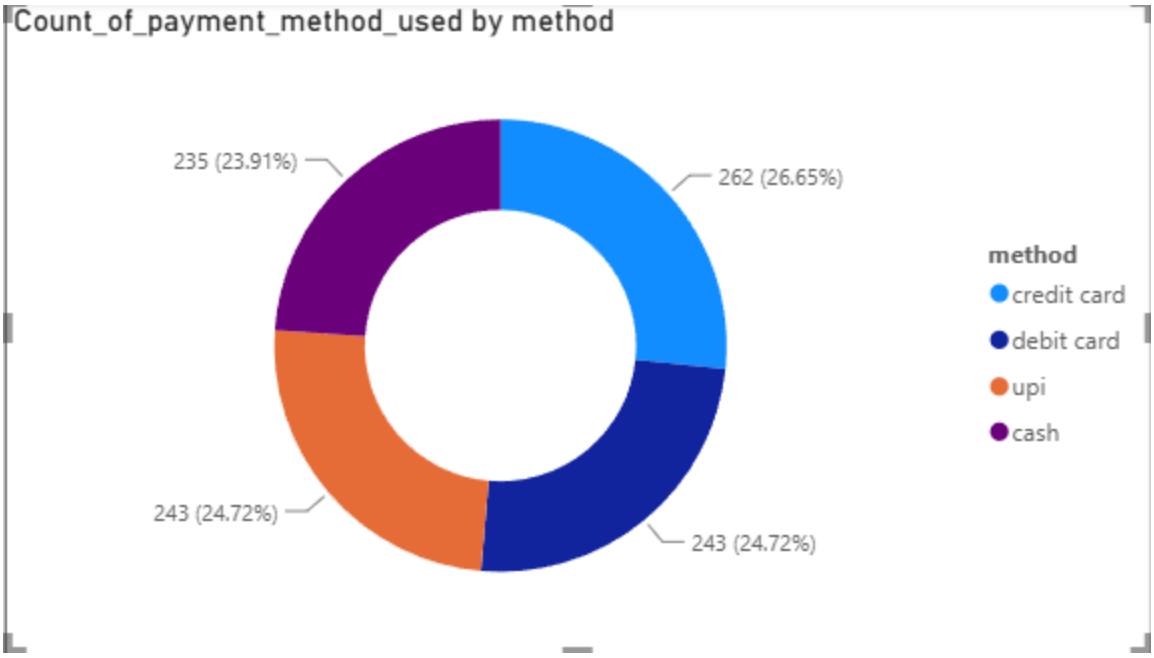
One interesting nuance: revenue doesn't move in perfect sync with trip count. For example, **16-17** has only 34 trips (one of the lowest counts) but its average fare per trip is the highest at ₹899 — suggesting longer or more expensive rides during that hour. Similarly, **8-9** has low trip count *and* the lowest average fare (₹641), making it a doubly weak revenue window.

Overall, the relationship between trip hour and revenue is **positively correlated with trip volume**, but average fare per trip adds a secondary layer of variation — hours like 0-1, 6-7, and 12-13 combine decent volume with higher average fares, making them the strongest revenue periods.

2.5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:



- **2 (UPI)** — 262 trips (26.65%) — most popular
- **1 (Cash)** — 243 trips (24.72%)
- **3 (Debit Card)** — 243 trips (24.72%)
- **4 (Credit Card)** — 235 trips (23.91%) — least popular

Payment methods are distributed almost evenly across all four options, with UPI being the most used at 262 trips (26.65%), followed by Cash and Debit Card both at 243 trips (24.72% each), and Credit Card the least used at 235 trips (23.91%).

The near-equal split across all four methods suggests customers don't have a strong preference for any single payment mode. However, UPI leading slightly — even if marginally — is consistent with the broader trend of digital payment adoption in urban India. Cash still holding nearly a quarter of trips indicates a significant portion of users either prefer or rely on offline payment.

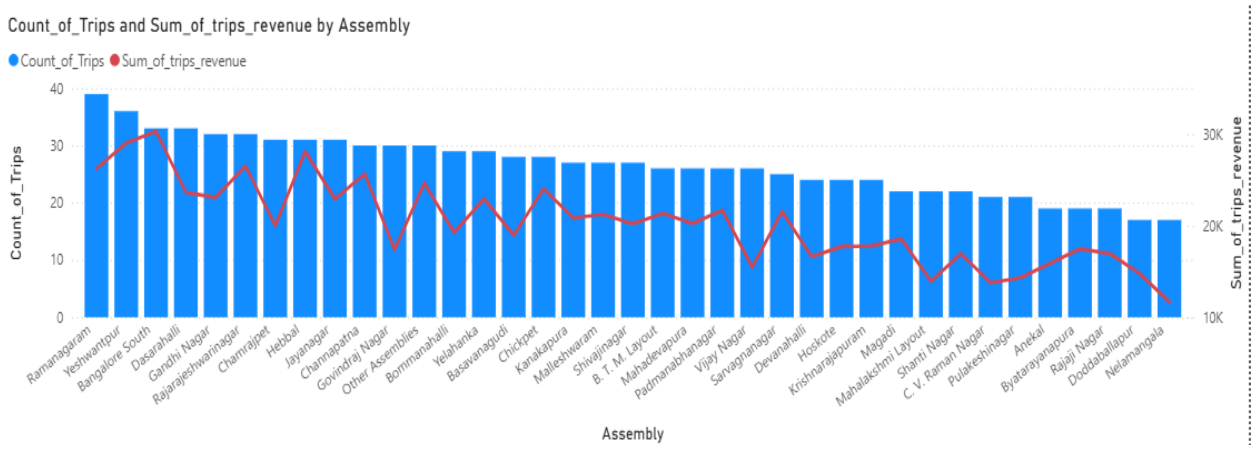
The almost identical share between Debit Card and Cash is also notable — it suggests Namma Yatri's customer base is fairly mixed in terms of digital readiness, which has implications for any cashless-only promotions or incentives the platform might consider.

2.6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation. Analyse factors contributing to their performance:

- **2.6.1. Rides:** Identify pickup zones with the highest number of trip requests.
  [3 marks]

Solution:



Among all 37 pickup zones, **Ramanagaram** leads with the highest number of trip requests at 39, followed by **Yeshwantpur** (36), and **Bangalore South** and **Dasarahalli** tied at 33 each. Gandhi Nagar and Rajarajeshwarinagar follow at 32 trips each, with Chamrajpet, Jayanagar and Hebbal at 31, and Govindraj Nagar rounding out the top 10 at 30 trips.

Ramanagaram topping the list is notable given it's a peripheral town rather than a central Bangalore zone — likely driven by outbound commuter demand. Yeshwantpur and Bangalore South are major transit and commercial hubs, which naturally attracts higher pickup volumes. The relatively narrow gap between the top 10 zones (39 down to 30) suggests demand is fairly distributed rather than concentrated in just one or two areas.

- **2.6.2. Revenue**: Identify pickup zones generating the highest revenue.
  [3 marks]

Solution:

When ranked by total fare generated, the picture shifts slightly from the ride count ranking. **Ramanagaram** still leads at ₹26,225 but **Bangalore South** jumps ahead of Dasarahalli despite having the same trip count (33), generating ₹30,295 against Dasarahalli's ₹23,624 — indicating longer or higher-fare routes from that zone. **Yeshwantpur** holds strong at ₹29,035, and **Hebbal** (₹28,097) performs better on revenue than its trip count rank would suggest.

The key insight here is that ride volume and revenue don't always move together — zones like Bangalore South and Hebbal generate disproportionately higher revenue relative to their trip count, pointing to longer distances or more expensive routes, while zones like Ramanagaram and Govindraj Nagar show the opposite pattern despite high pickup volumes.

2.7. Analyse Ride Time Periods Across Zones [4 Marks]

- Compare the trip trends for different time periods across pickup zones.

Solution:

Assembly	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22	23	T
Anekal	1	1	1	1	1		1	2		3		2	1	1				2			1	1			19
B. T. M. Layout		1	1		1	1	4		1	1	2	1		1			1	2	3	2	2	2			26
Bangalore South	1	1	1		1		1	1	1	1	1	2	2	4	1	2	2	2	3	1	1	1	2	1	33
Basavanagudi	2	2		2		2	7	1	1	1	2			1		1		1	1	2			2		28
Bommanahalli	2	2		1		2		1			3	2	1		3	1		4	1	1		3	1	1	29
Byatarayanapura	1			1	1				2	1		1		3	3	1		1	1	1			2		19
C. V. Raman Nagar		1		3		2	1	3	1	1		1	1		1			1				2	3		21
Chamrajpet	3	1	1	1	1	1	1	1			3	5	3	1	1	1	1			1		1	1	2	31
Channapatna	1	2	2	1	3	2	2	1	1	1	1		1	1			2	1	1	2	1	1	2	1	30
Chickpet	2		3				3	1		2		3	2	2		2		2		2		2	1	1	28
Dasarahalli	4	1		2	1		1		1	5		4	1	1	3	1	1	3	1	1	1		1		33
Devanahalli	1	1	1	1		2		2	1	1		1		3		3		1			1	4	1		24
Doddaballapur	2		3			2			1		1		1	1		1					1	2	1	1	17
Gandhi Nagar	3	2	1			1	3			2	1	1	2	1	2	3		3	2	1	1		2	1	32
Govindraj Nagar		1		2				2	1	1		1	2	2	1	2	4		1	1	1	3	1	4	30
Hebbal		2	1		1	1	1	1			2	2	3	1	1	3	2	1	2	4	1		1	1	31
Hoskote	3		2		1	1	3			2		1		1	2	2			1	3		1	1		24
Jayanagar	2	5		2		4	2	2		2				3	1		1	2	1		1	1	1	1	31
Kanakapura	1		1	2	1			2	3	1	3	1	3	2	1	1	1	1			1	1	1		27
Krishnarajapuram	2	1	2	2			2	1	3	1		1		1	1		1	1	2	1	2				24
Magadi		1	1	1	1	1				2	1		2	3				3	1		2		3		22
Mahadevapura	1		1	1	1	2	2	1	1	2		1	2	3	2				1			2	1	2	26
Mahalakshmi Layout	2		3		1	1	1		1	2	2	1	1		2			2	2				1		22
Malleshwaram	1	1	1		2	2			1		3	3		1	2	3	2	2		1		1	1		27
Nelamangala	2		2		2					1	1	1				3	2	2	1						17
Other Assemblies	2		1			1		1	2		4	1	2	2	2	1	1		1	1	1	2	4	1	30
Padmanabhanagar	3	1		1	2	2	2			2	1		1	2	2	2				3	1			1	26
Pulakeshinagar		1	1	2	1	1	2	1		1	2	2		1	1							3	2		21
Rajaji Nagar		1	1	1	3		1	3					2			1	1		2	1	1	1			19
Rajarajeshwarinagar	1	2	2		3	1		1	3			1	3	1	1	2	2		2		3	1	2	1	32
Ramanagaram	2	1	1	2	1	3	1	3	4	1					1		4	1	1	1	3	3	1	5	39
Sarvagnanagar	3	2	1	1	1	1	1	1			2	1			1	1	2	3	2				1	1	25
Shanti Nagar	1	3	1			1	1	2		2		1		2	2			1	1	1	1			1	22
Shivajinagar	1		1	1	1	1	1	1	1		1	2	3	3			1	2	3	1	1		2		27
Vijay Nagar	1		2	1		1	1	3		2			4	3	1		1	1		2		1	1	1	26
Yelahanka	2	1	1	2	1	1	3		1	1	2			1		1	1	1	3	1	3	1	2		29
Yeshwantpur		1	1	2	1	2		1	1	1	1	5	2		1	2	1	2	4		1	2	2	3	36
Total	53	39	41	36	33	42	48	39	33	43	39	48	45	52	39	40	34	48	44	35	32	40	48	32	983

To compare trip trends across pickup zones and time periods, a heatmap matrix was used with pickup zones as rows, duration buckets (0-1 through 23-24) as columns, and trip count as values with conditional color formatting — darker green indicating higher demand.

The most striking observation is **Basavanagudi during 6-7**, recording the highest single cell value of 7 trips — a clear early morning surge specific to that zone, likely driven by commuter traffic heading out before peak congestion. **Jayanagar (1-2)** and **Chamrajpet (11-12)** both hit 5 trips in their respective slots, making them the next most concentrated demand points.

Looking across the columns, the **0-1 hour bucket** has the highest column total (53) and shows activity spread across almost every zone — confirming the early morning demand pattern observed in 2.2, but now showing it isn't driven by just one or two zones; it's a platform-wide pattern.

Most zone-hour combinations sit at just 1-2 trips, meaning demand is fairly thin and spread out when broken down to this level of granularity. The darker cells are the exception rather than the rule, which suggests that while overall demand is distributed across zones, specific zone-hour combinations do have predictable spikes that could

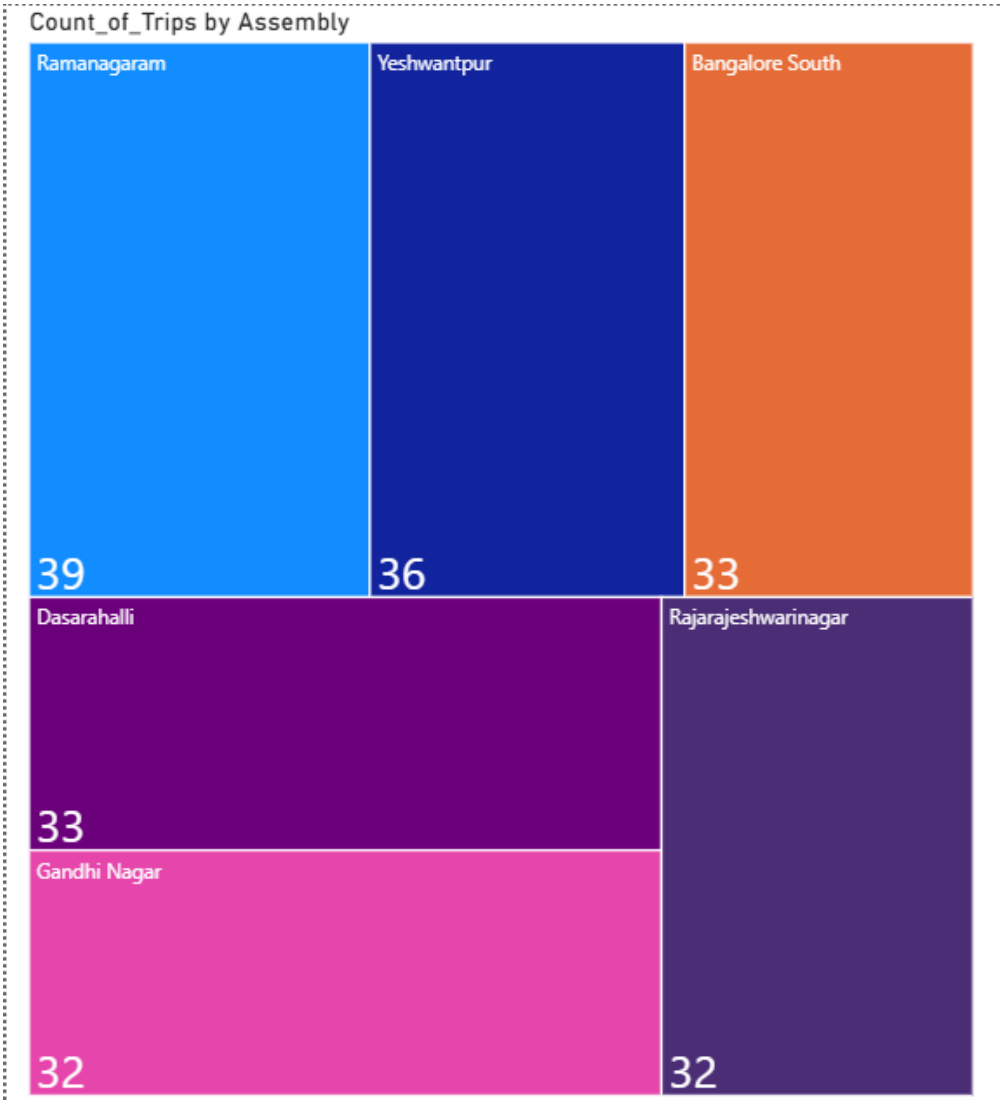
be used to strategically position drivers in advance.

Zones like **Ramanagaram, Sarvagnanagar, and Gandhi Nagar** show relatively consistent coloring across multiple time buckets rather than one sharp peak — indicating steady all-day demand rather than time-specific surges, which makes them reliable zones for consistent driver deployment.

2.8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

Solution:



A treemap was used to visualise the top zones by trip volume, where the size of each block directly represents the number of completed trips — making ranking comparisons immediately intuitive.

The top 5 filter returned **6 zones** due to ties at the lower ranks. **Ramanagaram** leads with 39 trips, followed by **Yeshwantpur** (36). **Bangalore South** and **Dasarahalli** are tied at 33 trips each, and **Gandhi Nagar** and **Rajarajeshwarinagar** are tied at 32 — hence all six are included for completeness.

Factors contributing to higher trip volumes in these zones:

- **Ramanagaram** — a satellite town with strong outbound commuter demand toward Bangalore, likely driving consistent pickup requests throughout the day
- **Yeshwantpur** — a major commercial and wholesale market hub with a busy railway station, naturally generating high cab demand across multiple time periods
- **Bangalore South** — a densely populated residential and commercial corridor with high daily mobility needs
- **Dasarahalli** — an industrial and residential zone on the northwestern fringe, with workers commuting in and out regularly
- **Gandhi Nagar and Rajarajeshwarinagar** — both are well-connected urban zones with a mix of residential, commercial, and transit activity driving steady demand

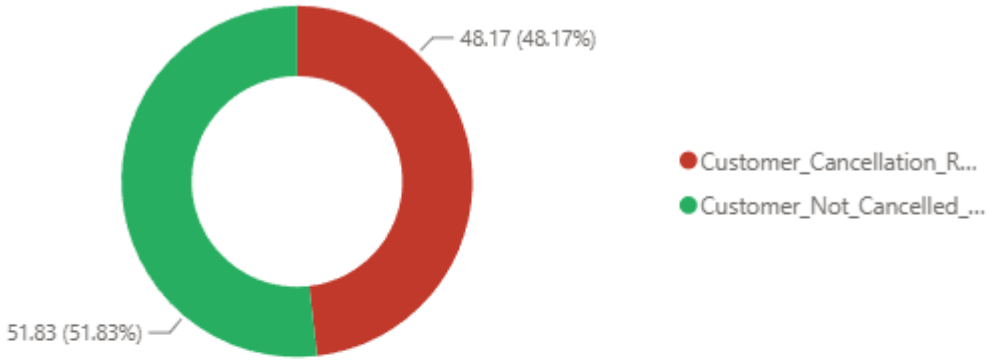
The common thread across all top zones is a mix of **transit connectivity, commercial activity, or high residential density** — zones that combine two or more of these factors consistently generate the highest trip volumes on the platform.

2.9. Basic Analytical Tasks [8 Marks]

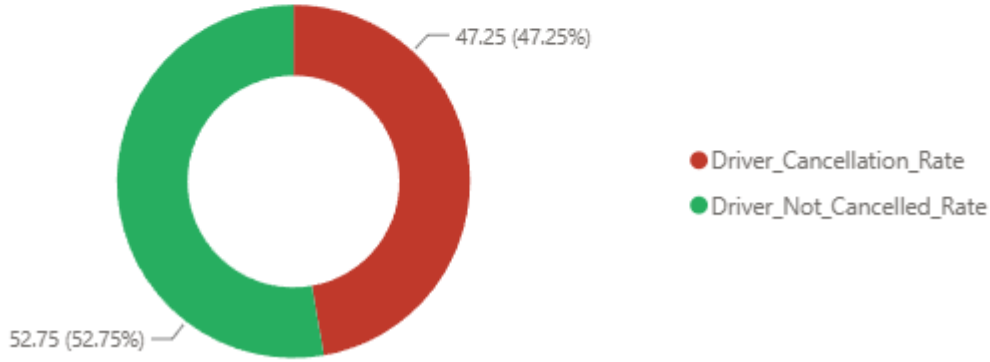
- **2.9.1**
  What are the percentages of cancellations and successful rides by both driver and customer? **[3 marks]**

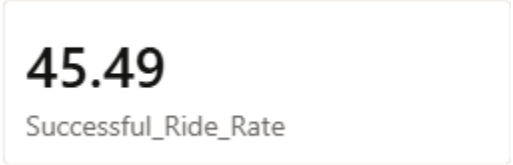
Solution:

Customer_Cancellation_Rate and Customer_Not_Cancelled_Rate



Driver_Cancellation_Rate and Driver_Not_Cancelled_Rate





Three visuals were used — two donut charts (customer and driver cancellations) and a card for overall successful ride rate.

Customer cancellations: 48.17% of trips saw a customer cancellation, while 51.83% proceeded without the customer cancelling. This is a notably high cancellation rate from the customer side — nearly half of all trip attempts were abandoned by the customer before completion.

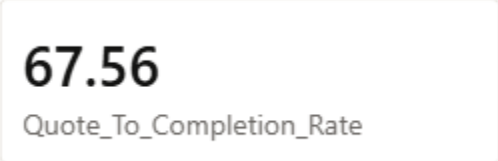
Driver cancellations: 47.25% of trips involved a driver cancellation, with 52.75% where the driver followed through. Driver cancellation rate is slightly lower than customer cancellation rate but still alarmingly close to 50%, suggesting drivers are also frequently dropping trip requests.

Successful ride rate: Only **45.49%** of all trip attempts in Trip_Details resulted in a completed ride — meaning more than half of all search attempts never converted into an actual trip. This is a significant operational concern and points to either a demand-supply mismatch (customers searching but not finding acceptable options) or a trust/reliability issue on both sides of the platform.

The combined picture suggests that cancellations — both customer and driver — are the single biggest barrier to platform efficiency, and addressing this should be a priority recommendation in the conclusion section.

- **2.9.2**
  Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods. **[5 marks]**

Solution:



Out of 2,161 total trip search attempts recorded in Trip_Details, **1,455 searches (67.33%)** progressed to the quote search stage. Of those who searched for quotes, **983 completed a ride — giving a quote-to-completion rate of 67.56%**.

Overall however, only **45.49%** of all search attempts converted into a completed ride, meaning roughly 1 in 2 searches never resulted in an actual trip. This gap between the quote-search rate (67.33%) and the final completion rate (45.49%) highlights that

a significant portion of users drop off somewhere between searching for quotes and actually completing the ride — likely at the cancellation stage, as confirmed in 2.9.1 where both customer and driver cancellation rates hover close to 48%.

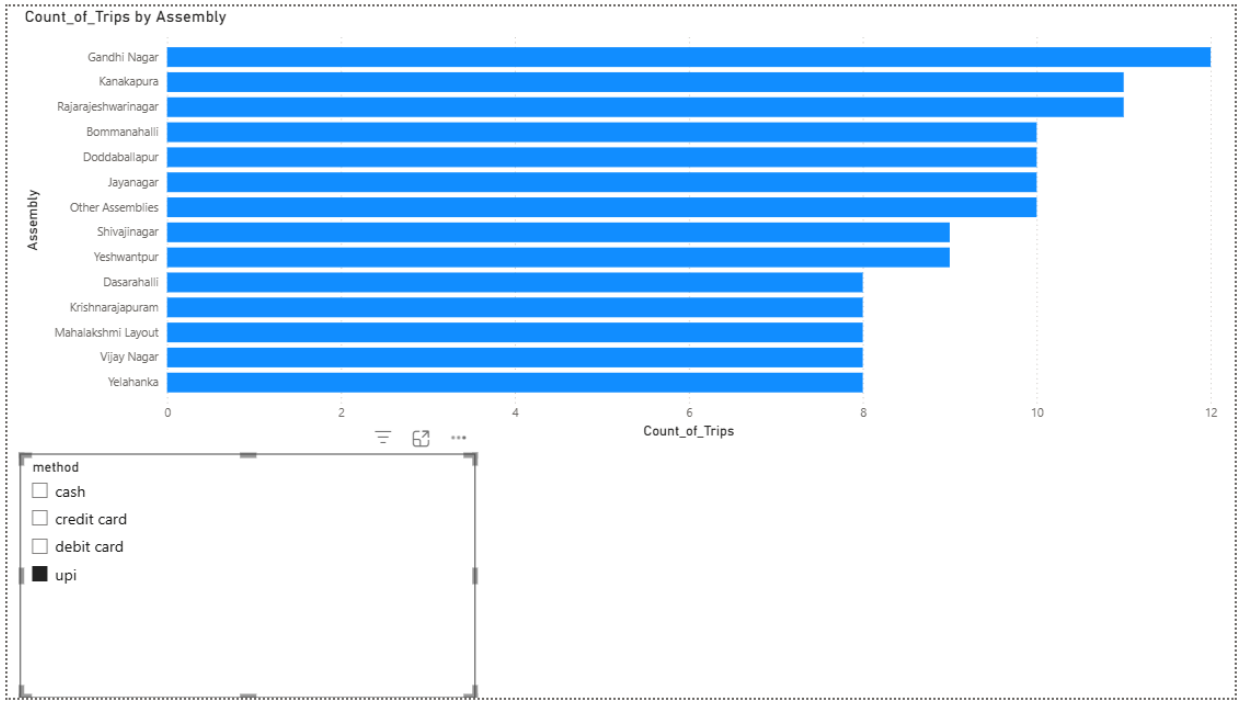
Regarding time-period variation — the dataset structure doesn't allow a direct breakdown of the quote-to-completion rate by hour, since Trip_Details (which contains the funnel flags) doesn't carry a time/duration column of its own. Duration is only available in the completed Trips table, meaning any time-based filter automatically restricts the context to completed rides only, producing a flat 100% rate across all hours — which is a structural limitation of the data rather than a meaningful insight.

The overall funnel tells the clearest story here: of every 100 people who searched, 67 looked for quotes, and only 45 actually completed a ride — making cancellation reduction the single most impactful lever for improving platform conversion.

2.10. Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

Solution:



Filtered on payment method used as upi in the respective visualization.

A payment method slicer was created as an interactive parameter filter, allowing dynamic filtering of trip distribution across pickup zones by payment type — cash, UPI, debit card, and credit card. This filter was chosen because payment method is a meaningful categorical dimension that cuts across all zone and trip data, enabling comparison of customer payment preferences zone by zone without building separate visuals for each method.

Insights drawn:

When no filter is applied, Ramanagaram leads overall with 39 trips across all payment methods, consistent with findings in 2.6 and 2.8. Selecting individual payment methods reveals how trip volume redistributes across zones depending on the payment type chosen — some zones maintain their top ranking regardless of payment method while others drop significantly, suggesting that certain zones have a stronger preference for specific payment modes.

The near-equal split across all four payment methods observed in 2.5 (cash 24.72%, UPI 26.65%, debit card 24.72%, credit card 23.91%) means that filtering by any single method roughly reduces each zone's trip count to approximately one quarter of its total — confirming no single payment method dominates any particular zone strongly enough to distort the overall zone rankings.

This parameter is particularly useful for targeted marketing — for example, if Namma Yatri wanted to run a UPI cashback promotion, filtering by UPI immediately shows which zones have the highest UPI-paying customer base, allowing geographically targeted campaigns rather than a blanket approach.

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3. Conclusion [20 Marks]

3.1. Recommendations for Operational Efficiency [10 Marks]

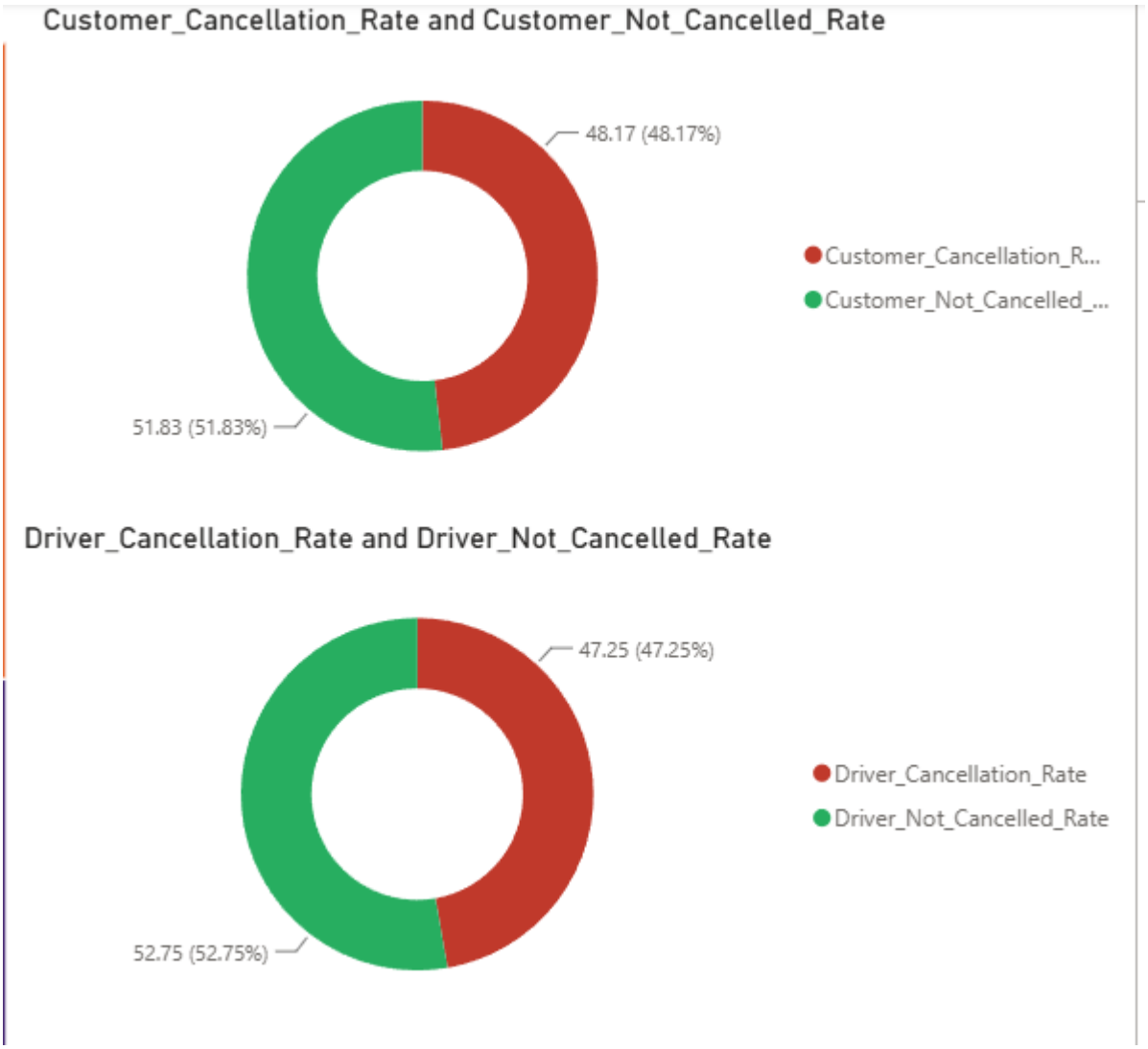
- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

Solution:

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1. Cancellation Reduction — Highest Priority

The most critical operational issue identified is the near-50% cancellation rate on both sides — customer cancellations at 48.17% and driver cancellations at 47.25%, resulting in only 45.49% of all search attempts converting into completed rides. This means more than half the platform's demand is being lost before a ride is even completed.



Recommendations:

- Introduce a **cancellation penalty system** for both drivers and customers after a threshold number of cancellations within a time window
- Show customers **real-time driver location** before booking confirmation to reduce uncertainty-driven cancellations

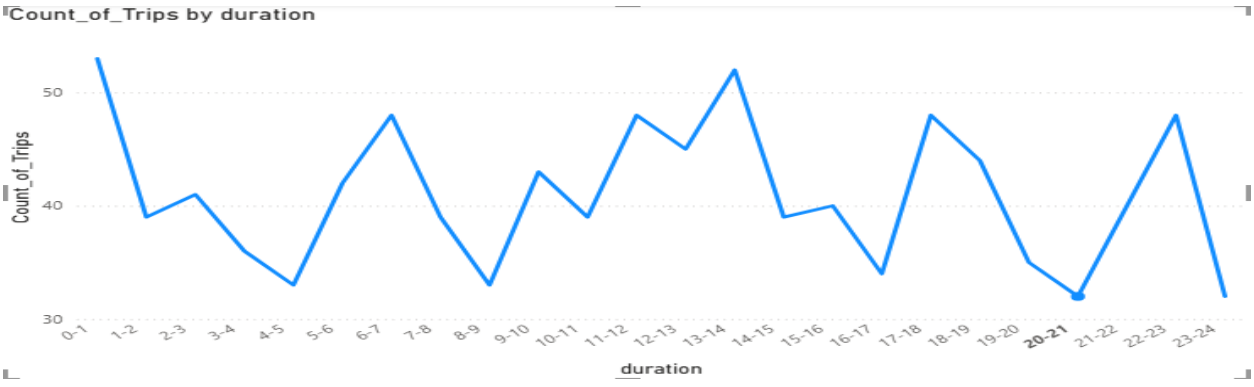
- Implement **driver incentives** for maintaining low cancellation rates — bonus payments or priority ride allocation for high-completion drivers

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2. Driver Deployment by Hour and Zone

From the demand analysis (2.2) and heatmap (2.7), demand peaks at 0-1, 6-7, 11-12, 13-14 and 17-18, and is concentrated in zones like Ramanagaram, Yeshwantpur and Bangalore South.

Assembly	0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22	2	T
Anekal	1	1	1	1	1		1	2		3		2	1	1				2			1	1			19
B. T. M. Layout		1	1		1	1	4		1	1	2	1		1			1	2	3	2	2	2			26
Bangalore South	1	1	1		1		1	1	1	1	1	2	2	4	1	2	2	2	3	1	1	1	2	1	33
Basavanagudi	2	2		2		2	7	1	1	1	2			1		1		1	1	2			2		28
Bommanahalli	2	2		1		2		1			3	2	1		3	1		4	1	1		3	1	1	29
Byatarayanapura	1			1	1				2	1		1		3	3	1		1	1	1			2		19
C. V. Raman Nagar		1		3		2	1	3	1	1		1	1		1			1				2	3		21
Chamrajpet	3	1	1	1	1	1	1	1	1		3	5	3	1	1	1	1			1		1	1	2	31
Channapatna	1	2	2	1	3	2	2	1	1	1	1		1	1			2	1	1	2	1	1	2	1	30
Chickpet	2		3				3	1		2		3	2	2		2		2		2		2	1	1	28
Dasarahalli	4	1		2	1		1		1	5		4	1	1	3	1	1	3	1	1	1		1		33
Devanahalli	1	1	1	1		2		2	1	1		1		3		3		1			1	4	1		24
Doddaballapur	2		3			2			1		1		1	1		1					1	2	1	1	17
Gandhi Nagar	3	2	1			1	3			2	1	1	2	1	2	3		3	2	1	1		2	1	32
Govindraj Nagar		1		2				2	1	1		1	2	2	1	2	4		1	1	1	3	1	4	30
Hebbal		2	1		1	1	1	1			2	2	3	1	1	3	2	1	2	4	1		1	1	31
Hoskote	3		2		1	1	3			2		1		1	2	2			1	3		1	1		24
Jayanagar	2	5		2		4	2	2		2				3	1		1	2	1		1	1	1	1	31
Kanakapura	1		1	2	1			2	3	1	3	1	3	2	1	1	1	1			1	1	1		27
Krishnarajapuram	2	1	2	2			2	1	3	1		1		1	1		1	1	2	1	2				24
Magadi		1	1	1	1	1				2	1		2	3				3	1		2		3		22
Mahadevapura	1		1	1	1	2	2	1	1	2		1	2	3	2				1			2	1	2	26
Mahalakshmi Layout	2		3		1	1	1		1	2	2	1	1		2			2	2				1		22
Malleshwaram	1	1	1		2	2			1		3	3		1	2	3	2	2		1		1	1		27
Nelamangala	2		2		2					1	1					3	2	2	1						17
Other Assemblies	2		1			1		1	2		4	1	2	2	2	1	1		1	1	1	2	4	1	30
Padmanabhanagar	3	1		1	2	2	2			2	1		1	2	2	2				3	1			1	26
Pulakeshinagar		1	1	2	1	1	2	1		1	2	2		1	1								3	2	21
Rajaji Nagar		1	1	1	3		1	3					2			1	1		2	1	1	1			19
Rajarajeshwarinagar	1	2	2		3	1		1	3			1	3	1	1	2	2		2		3	1	2	1	32
Ramanagaram	2	1	1	2	1	3	1	3	4	1					1		4	1	1	1	3	3	1	5	39
Sarvagnanagar	3	2	1	1	1	1	1	1			2	1			1	1	2	3	2				1	1	25
Shanti Nagar	1	3	1			1	1	2		2		1		2	2			1	1	1	1	1		1	22
Shivajinagar	1		1	1	1	1	1	1	1		1	2	3	3			1	2	3	1	1		2		27
Vijay Nagar	1		2	1		1	1	3		2			4	3	1		1	1		2		1	1	1	26
Yelahanka	2	1	1	2	1	1	3		1	1	2			1		1	1	1	3	1	3	1	2		29
Yeshwantpur		1	1	2	1	2		1	1	1	1	5	2		1	2	1	2	4		1	2	2	3	36
Total	53	39	41	36	33	42	48	39	33	43	39	48	45	52	39	40	34	48	44	35	32	40	48	32	983



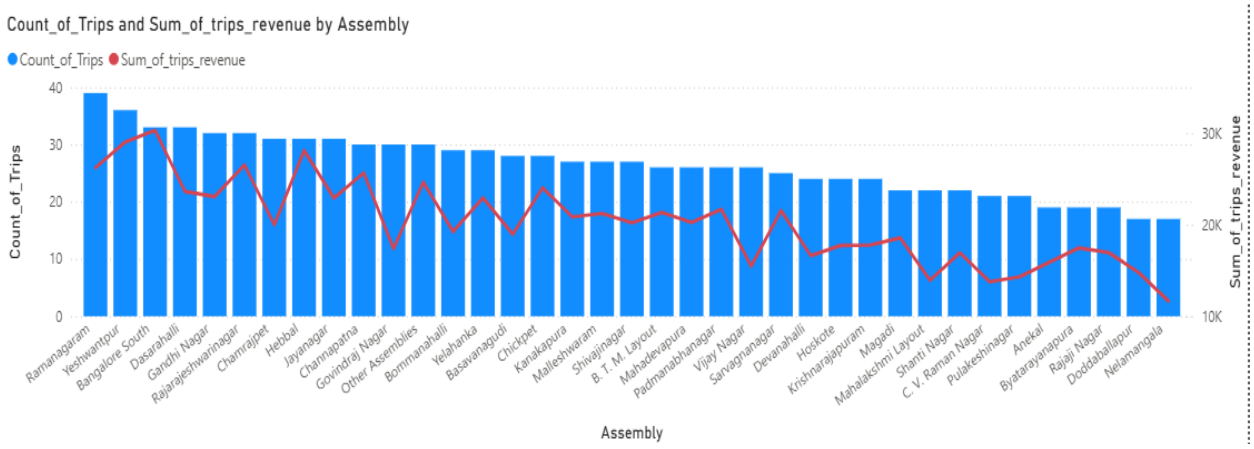
Recommendations:

- Use **predictive driver positioning** — push notifications to drivers in high-demand zones during peak hours identified in the heatmap
- Reduce driver idle time during low-demand windows (8-9, 20-21, 23-24) by incentivising drivers to reposition to high-demand zones proactively

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3. Revenue Optimisation by Zone

From 2.6.2, zones like Bangalore South and Hebbal generate disproportionately higher revenue relative to their trip count, suggesting longer or higher-fare routes. Conversely, Ramanagaram has the highest trip count but relatively lower revenue per trip.



Recommendations:

- Apply **dynamic pricing** during peak hours in high-revenue zones to maximise fare per trip
- Investigate whether shorter trips in Ramanagaram can be supplemented with incentives for drivers to accept longer routes, improving revenue per trip in that zone

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4. Quote-to-Completion Funnel

Only 67.33% of searches progressed to quote search stage, and of those, 67.56% completed a ride. The biggest dropout happens between searching and completing — largely driven by cancellations.

Recommendations:

- Streamline the **quote acceptance flow** — reduce the number of steps between receiving a quote and confirming a booking to minimise drop-off
- Introduce **auto-acceptance options** for frequent customers who repeatedly book similar routes

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3.2. Marketing and Operational Strategy Improvements [10 Marks]

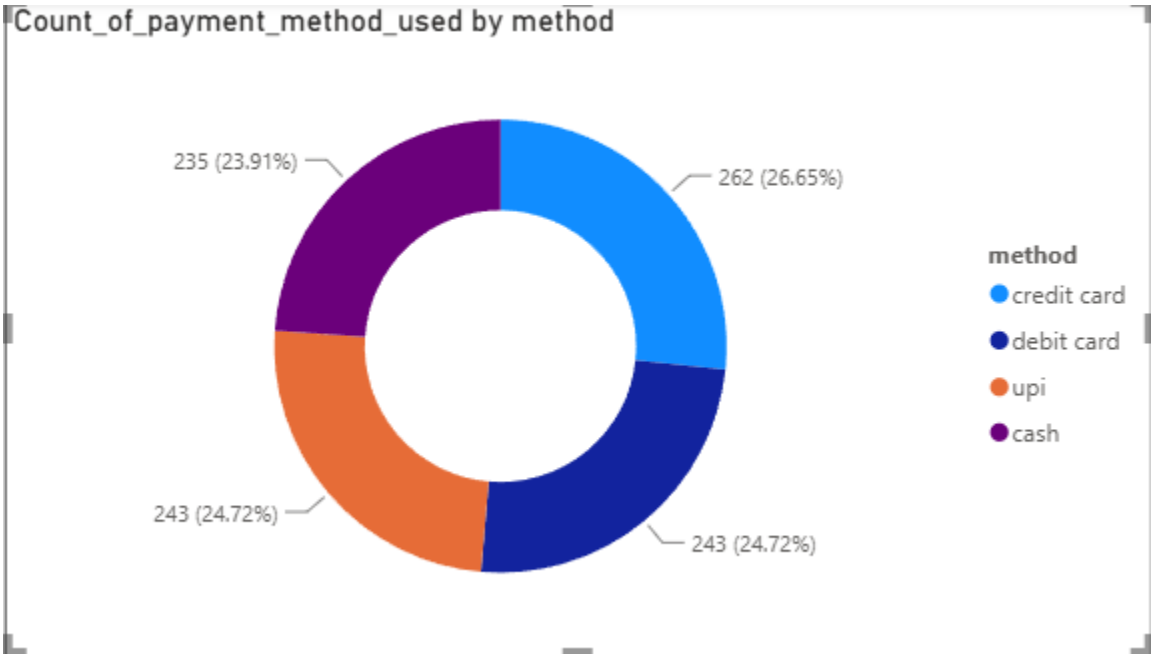
- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

Solution:

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1. Payment Method Promotions

From 2.5, all four payment methods are almost equally distributed (cash 24.72%, UPI 26.65%, debit card 24.72%, credit card 23.91%), suggesting no strong digital payment preference has emerged yet despite UPI being marginally ahead.



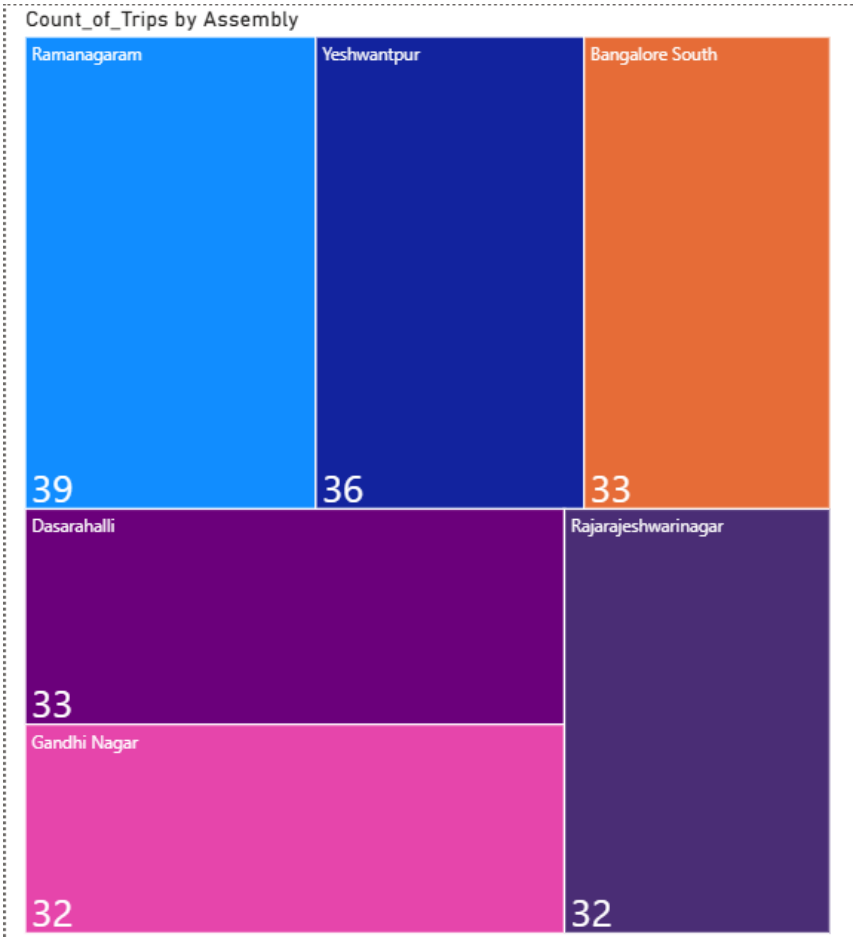
Recommendations:

- Run **UPI cashback promotions** to nudge the remaining ~75% of non-UPI users toward digital payments — reduces cash handling friction for drivers and speeds up trip completion
- Partner with banks for **debit/credit card ride discounts** during off-peak hours (8-9, 20-21) to stimulate demand during the platform's weakest windows
- Use the payment method slicer analysis (2.10) to identify which zones have higher cash dependency and target those zones specifically with digital payment awareness campaigns

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2. Zone-Specific Marketing

From 2.6 and 2.8, the top performing zones (Ramanagaram, Yeshwantpur, Bangalore South) already generate strong demand organically. The bottom zones (Nelamangala, Doddaballapur, Rajaji Nagar) show consistently low trips and revenue.



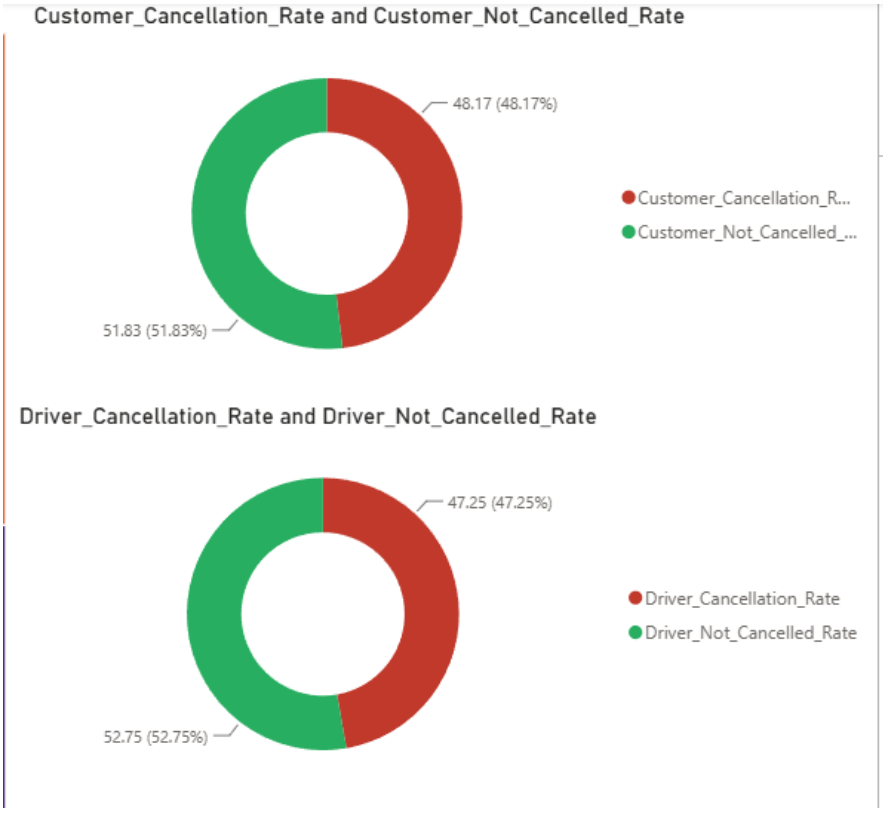
Recommendations:

- **Defend top zones** — loyalty rewards for frequent riders in Ramanagaram and Yeshwantpur to retain existing high-demand customer base
- **Grow bottom zones** — targeted first-ride discount campaigns in Nelamangala and Doddaballapur to build awareness and trial in underperforming areas
- Position Namma Yatri as the **commuter app of choice** in peripheral zones like Ramanagaram where outbound commuter demand is already naturally high

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3. Driver Incentive Programs

From 2.9.1, driver cancellation rate (47.25%) is almost as high as customer cancellation rate (48.17%), suggesting drivers are selectively rejecting or abandoning trips at a damaging rate.



Recommendations:

- Introduce a **driver rating and reward system** — drivers maintaining cancellation rates below 20% receive priority ride allocation and weekly bonuses
- Offer **surge pricing during peak hours** (0-1, 6-7, 13-14) identified in 2.2 to

financially incentivise drivers to be online and accepting rides during high-demand windows

- Provide **guaranteed minimum earnings** per hour during off-peak slots to keep driver supply available even when demand is lower

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4. Customer Retention Strategy

With only 45.49% of searches converting to completed rides, a large portion of potential customers are experiencing failed bookings and may be switching to competitor platforms.

Recommendations:

- Introduce a **"ride guarantee" feature** — if a booking is cancelled by a driver, the customer is automatically rematched within 2 minutes rather than restarting the search
- Send **re-engagement notifications** to customers who searched but didn't complete a ride, offering a small discount on their next booking to bring them back
- Build a **frequent rider program** rewarding customers who complete a high number of rides monthly — converting occasional users into loyal, habitual riders

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